Ma Haojie

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Personal Information

Male, Born in 1994 Position: Algorithm Engineer Work Experience: ~5 years

Skills: 3D Object Detection, Reinforcement Learning (DQN, PPO, GRPO, GSPO), VLM, LLM fine-tuning and distillation, c++, cuda.

Education

- Master, Nanjing University, Computer Science and Technology, 2017.9~2020.7
- Bachelor, Nanjing University, Computer Science and Technology, 2013.9~2017.7

Work Experience

- **DeepRoute, Perception Algorithm Engineer, 2023.1~Now**
- **Pony.ai, Perception Algorithm Engineer, 2021.8~2022.12**
- **Microsoft China, m365 Department, SDE, 2020.6~2021.7**

Research Experience (Total Citations 350+)

- AttnSense: Multi-level Attention Mechanism For Multimodal Human Activity Recognition (**Ijcai 2019, CCF A**, attention-based multimodal fusion)
- Unsupervised Human Activity Representation Learning with Multi-task Deep Clustering (**UbiComp 2020, CCF A**, unsupervised pre-training related)

Project Experience

Project 1: End-to-End Autonomous Driving Large Model Based on VLA Architecture (DeepRoute)

- Project Background: Build a unified multimodal autonomous driving large model, aiming to achieve an end-to-end closed loop from perception understanding to decision planning. Construct a unified model with scene understanding, trajectory output, and risk assessment capabilities.
- Technical Details:
  - Model Architecture
    - Base model adopts Qwen2-VL 7B architecture, overall parameter count is 8B, precision is FP8.
    - Input includes multimodal information such as BEV features, navigation maps, perception maps, etc., outputs include scene descriptions, trajectory planning, speed limit information, longitudinal/lateral planning, risk flags. Introduces COT output design, requiring the model to first output scene semantic descriptions and intent predictions before generating trajectories.
  - Pretraining
    - Built heterogeneous data mixed training, mixing general visual-language datasets with autonomous driving datasets (large-scale BEV-description pairs) for joint training.
    - Future consideration of combining Masked World Modeling (MWM) concepts to allow the model to learn physical laws from unannotated massive road test videos, improving the model's spatial perception capability.
  - Post-training: Adopted SFT supervised fine-tuning and GRPO reinforcement learning training, optimizing model performance in game scenarios through multi-dimensional reward functions. Reward functions include:
    - Format Reward: Verify the completeness of output structure (scene description, trajectory, speed limit, risk flag labels, etc.)
    - Scene Label Reward: Assess scene understanding accuracy (multi-label classification)
    - Trajectory Reward: Evaluate reference line trajectory generation quality (matching degree with ground truth)
    - Speed Reward: Assess speed token prediction accuracy
    - Risk Flag Reward: Assess safety decision-making capability (F0.5 score evaluation)
    - Diversity Reward: Promote model adaptability in corner case scenarios
  - Data Aspect:
    - Deployed VLM distillation model on vehicle for scene classification, performing corner case collection, including irregular obstacles, extreme road structures (narrow roads, roundabouts), interactive gaming (forced lane cutting, unprotected left turns).
- Delivery Metrics:
  - Improved complex urban scenario disengagement distance (MPI) by 40%.
  - Increased success rate for typical Corner Cases (such as unprotected left turns) from 75% to 92%.
- Current Issues
  - Model generalization capability is insufficient with obvious overfitting

    - Trajectory output is unreasonable, still requiring significant rule-based fallback downstream

Project 2: 3D Perception and OCC Perception (DeepRoute)

- QueryTracker
    - Project Background: Use neural network-based tracker to replace the original rule-based tracker
    - Adopted ideas similar to TrackFormer and TransCenter, based on query Transformer architecture
        - Used Instance Bank to store object historical trajectories, which is key to QueryTracker achieving stable long-term tracking, solving problems like target occlusion, brief disappearance, etc., ensuring tracking continuity and stability.
        - Model inputs current BEV features, historical BEV features, bbox, confidence, time intervals, pose and other information, directly outputs tracked 3D obstacles through self-attention and cross-attention
- 3D OCC
    - Project Background: OCC recognition based on 3D perception, used to identify different types of occupancy objects (water barriers, fences, cones, etc.)
    - Model Architecture:
        - Fully convolutional head, used for 3D segmentation, judging whether occupied, occupied object categories and other information
        - Addressed rain false detection issues by adding noise class prediction
        - Explored fusion OCC, single LIDAR OCC encountered bottlenecks, combined image data to implement fusion OCC.

Project 3: Point Cloud Detection and Segmentation (Pony.ai)

- Project Description: Reproduced and optimized the top-tier paper Range Sparse Net (RSN) from Waymo, building an end-to-end perception framework that balances Range View (RV) semantic segmentation and BEV object detection.
- Model Architecture:
    - Performed 3D semantic segmentation (similar to OCC) using U-Net in RV view, identifying noise, ground points, background points, pedestrians, vehicles and other information, with loss function using Focal Loss + Lovász-Softmax Loss
    - Object detection was performed in BEV view, with model architecture adopting Sparse CNN + SSD/CenterPoint Head, loss function using multitask loss containing classification loss (focal loss), regression loss (l1 loss), Direction Loss: specifically designed to handle periodicity issues with orientation angles
    - Sparse convolution network input was point cloud filtered after Range View segmentation, with background points removed. The specific input format was spatial coordinates (x, y, z) of non-zero points. Sparse convolution was implemented via cuda, with the specific mechanism combining hash tables and Rule Book (input coordinates corresponding to convolution kernel output points). With the Rule Book, the calculation process becomes matrix multiplication (GEMM), making sparse convolution extremely fast on GPUs

Project 4: Data Lineage System Based on SCOPE Static Analysis (Microsoft)

- Project Description: Developed a static parsing engine targeting Microsoft's proprietary big data language SCOPE, solving data redundancy and asset tracing challenges on the Cosmos platform by constructing full-link data lineage graphs.
- Key Contributions:
    - Compiler-Level Parser Development: Based on compiler principles, independently implemented a parser for SCOPE syntax, extracting dependencies between data tables, views and UDFs by parsing Abstract Syntax Trees (AST).
    - Large-Scale Lineage Graph Construction: Processed and associated production and consumption records of PB-level data within the Cosmos platform, achieving table-level and field-level (Column-level) lineage tracking.
- Business Value:
    - Cost Optimization: Identified and cleaned "cold data" and redundant copies without downstream calls for 180 days based on lineage relationships, directly saving approximately 12% of storage costs for the department.
    - Security Compliance: Provided clear traceability paths for sensitive data, meeting internal data compliance and impact analysis requirements (EU GDPR).

Project 5: Cluster Monitoring System Based on Azure ML (Microsoft)

- Project Description: For Microsoft's ultra-large-scale computing cluster Cosmos, built a closed-loop system from anomaly monitoring to automated operations and maintenance, aimed at reducing cluster downtime risks and improving job stability.
- Key Contributions:
    - Multi-dimensional Metrics Monitoring: Integrated underlying hardware (CPU, disk I/O, network packet loss) and system-level fault logs of Cosmos servers, constructing a high-dimensional time-series feature database.
    - Anomaly Detection Architecture: Connected to Azure Time Series Anomaly Detection service, utilizing its built-in SR-CNN (Spectral Residual-CNN) algorithm to achieve real-time monitoring of non-stationary time series data. Compared to traditional threshold alerts, the false negative rate decreased by 18%.

马浩杰

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个人信息

男，1994 年出生 求职意向：算法工程师 工作经验：约5年

技能清单: 3D目标检测，强化学习(DQN, PPO, GRPO, GSPO)，VLM，LLM微调和蒸馏，c++, cuda.

教育经历

- 硕士，南京大学，计算机科学与技术专业，2017.9~2020.7
- 学士，南京大学，计算机科学与技术专业，2013.9~2017.7

工作经历

- **元戎启行，perception 算法工程师, 2023.1~至今**
- **小马智行，perception 算法工程师, 2021.8~2022.12**
- **微软中国 公司，m365 部门，SDE, 2020.6~2021.7**

科研经历(总引用量350+)

- AttnSense: Multi-level Attention Mechanism For Multimodal Human Activity Recognition (**Ijcai 2019, CCF A**, attention多模态融合)
- Unsupervised Human Activity Representation Learning with Multi-task Deep Clustering (**UbiComp 2020, CCF A**, 无监督预训练相关)

项目经历

项目一: 基于VLM架构的二段式端到端自动驾驶(元戎启行)

- 项目背景：构建统一的多模态自动驾驶大模型，旨在实现从感知理解到决策规划的端到端闭环, 构建具备场景理解，行为决策，风险评估能力的统一模型
- 技术细节：
  - 模型整体架构
    - 初版基座模型采用Qwen2-VL 7B架构，后续砍到0.5B.
    - 输入为BEV特征,导航地图，感知地图, agent轨迹等多模态信息，通过每个模态独有的encoder，编码为bev token, sdmap token, agent token等.
    - 输出为场景描述、轨迹规划、限速信息、纵横向规划、风险标志, 引入COT输出设计，要求模型在生成轨迹前先输出场景语义描述,关键障碍物与意图预测.
  - 预训练
    - 构建了异构数据混合训练，将通用视觉-语言数据集与自动驾驶数据集（大规模BEV-描述对, 描述对包含3d bbox, occ等描述对）进行混合训练
    - 下一代端到端系统开发，考虑结合Masked World Modeling (MWM) 思想，让模型在未标注的海量路测视频中学习物理规律，提高模型的空间感知能力, 联合分布输出未来frame和action.
  - 后训练：采用SFT监督微调和GRPO强化学习训练，通过多维度奖励函数优化模型在博弈场景下的表现。奖励函数包括：
    - 格式奖励：验证输出结构的完整性（场景描述、轨迹、限速、风险标志等标签）
    - 场景标签奖励：评估场景理解准确性（多标签分类）
    - 横纵向决策奖励：评估横纵向规划的准确度
    - 轨迹奖励：评估参考线轨迹生成质量（与ground truth的匹配度）
    - 速度奖励：评估速度token预测准确性
    - 风险标志奖励：评估安全决策能力（F0.5分数评估）
    - 多样性奖励：促进模型在corner case场景下的适应性
  - 数据方面:
    - 构建了video级别的感知和规控tag, 包含cutin, junction, lane change, overline等不同的驾驶场景.
    - 车端部署VLM蒸馏模型用作场景分类，进行corner case采集,包含异形障碍物，极端道路结构(窄路，环岛)，交互博弈（强行加塞, 无保护左转）
  - 车端应用， 和二段式planer构成典型的快慢系统，处理 90% 以上的常规驾驶, 当慢系统发现复杂场景(施工路段, 收费站etc通行等)时，会修改快系统的输入参数或直接接管决策

项目二: 3D感知与OCC感知(元戎启行)

- QueryTracker
  - 项目背景：使用基于神经网络的tracker替代原先的基于规则的tracker
  - 采用了类似于TrackFormer和TransCenter的思想，基于query的Transformer架构
    - 使用Instance Bank存储object历史轨迹,是QueryTracker实现稳定长期跟踪的关键，解决了目标遮挡、短暂消失等问题，确保跟踪的连续性和稳定性。

        - 模型输入当前BEV特征，历史的BEV特征，bbox, 置信度，时间间隔，位姿等信息，通过self attention和cross attention直接输出追踪后的3D障碍物
- 3D occ
    - 项目背景：基于3D感知的OCC识别，用于识别不同类型的动态和静态占用物（水马、围栏、锥桶等）
    - 模型架构：
        - 全卷积的head，用于做3D分割，判断是否占据, 占据物体类别等信息
        - 针对雨花误检问题，增加noise类别预测
        - 针对80-160m范围，添加远端occ head.
        - 动静态occ head实现，输出occ flow(vx, vy)

项目三：点云检测和分割(小马智行)

- 项目描述： 复现并优化 Waymo 顶刊论文 Range Sparse Net (RSN)，构建一套兼顾 Range View (RV) 语义分割与 BEV 目标检测的端到端感知框架。
- 模型架构:
    - 在RV视角使用U-Net进行3D语义分割(类似occ)，识别噪点，地面点，背景点，行人车辆等信息，损失函数使用Focal Loss + Lovász-Softmax Loss
    - 目标检测在BEV视角进行，模型架构采用Sparse CNN + SSD/CenterPoint Head, 损失函数采用多任务损失，包含分类损失(focal loss), 回归损失(l1 loss), Direction Loss：专门用于处理朝向角的周期性问题
    - 稀疏卷积网络输入为经过 Range View 分割过滤后的点云，背景点已被去除，输入的具体形式为非零点的空间坐标 (x, y, z), 稀疏卷积通过cuda实现，具体机制为结合哈希表和Rule Book(卷积核输出点对应的输入坐标),有了 Rule Book，计算过程就变成了矩阵乘法（GEMM），这让稀疏卷积在 GPU 上的执行速度飞快

项目四: 基于 SCOPE 静态分析的数据血缘系统(微软)

- 项目描述：开发了一套针对微软大数据特有语言 SCOPE 的静态解析引擎，通过构建全链路数据血缘图谱，解决 Cosmos 平台数据冗余及资产溯源难题
- 核心贡献：
    - 编译级 Parser 开发：基于编译原理，自主实现了一套针对 SCOPE 语法的解析器（Parser），通过解析 抽象语法树 (AST) 提取数据表、视图及 UDF 之间的依赖关系。
    - 大规模血缘图谱构建：处理并关联 Cosmos 平台内 PB 级数据的生产与消费记录，实现了表级与字段级（Column-level）的血缘追踪。
- 业务价值：
    - 成本优化：基于血缘关系识别并清理了长达 180 天无下游调用的“冷数据”和冗余副本，直接为部门节省了约 12% 的存储开销。
    - 安全合规：为敏感数据提供了清晰的溯源路径，满足了内部数据合规与影响分析的需求(欧盟GDPR法案)

项目五: 基于Azure ML的集群监控系统

- 项目描述：针对微软o365团队内部存储计算集群 Cosmos，搭建了一套从异常监测到自动化运维的闭环系统，旨在降低集群宕机风险并提升任务作业的稳定性。
- 核心贡献：
    - 多维指标监控：集成 Cosmos QPS，RT，失败率等, 服务器底层硬件（CPU、磁盘 I/O、网络丢包）与系统级故障日志，构建了高维时序特征库。
    - 异常检测架构：接入 Azure Time Series Anomaly Detection 服务，利用其内置的SR-CNN(Spectral Residual-CNN)+孤立森林算法检测，实现了对非平稳时序数据的实时监控。相较于传统阈值告警，漏报率降低了 18%